

SPRING BOOT & MICRO SERVICES INTERVIEW QUESTIONS

S.NO	Topic Name	Sub - Topics
1	Spring Core	Fundamentals of Spring Core Bean Lifecycle Spring Bean scopes (Singleton (default), Prototype, Request, Session, Global session) Bean Factory and Application Context Inversion of Control (IoC) Dependency Injection (DI), @Autowired(ByName, ByType, ByConstructor) Bean Configuration (XML Configuration, Java-Based Configuration, Annotation-Based Configuration) Environment & Properties, @Value annotations
2	Spring Boot core	Introduction to Spring Boot Spring Boot Project Setup Spring Initializer, Spring Boot Starters Auto Configuration @SpringBootApplication internals, SpringApplication.run() internals Application Properties & Configuration Profiles Embedded Servers Spring Boot DevTools Spring Boot Logging Spring Boot Actuator Exception Handling Externalized Configuration (Environment variables, Command-line arguments, Property precedence order)
3	Spring Data Jpa	JDBC Template JPA Overview, ORM Basics Hibernate overview Spring Data JPA Overview, Crud Repository, JPA Repository DB Operations (save(), saveAll(), findAll() etc..) Hikari CP Pool Entities, DTOs & DAO classes FindByXXX methods syntax Query By Example JPQL and Native Query, @Query annotations Pagination and Sorting Embedded DB Introduction (H2), Auditing with @CreateDate, @LastModifiedDate Relationships(one to one, one to many, many to one, many to many), @Transactional annotation, N+1 problem and how to solve it with @EntityGraph or JOIN FETCH
4	Spring MVC	Spring MVC Architecture (Model–View–Controller) Spring MVC Request flow (DispatcherServlet, HandlerMapper, ViewResolver) Spring MVC Configuration, Embedded Server (Tomcat, Jetty, Undertow) Spring MVC Controller, QueryParam, Path Param Spring MVC Validation Spring MVC Session Management, Thymeleaf Introductions
5	Spring Rest API	Distributed Application, SOAP Overview REST Overview API overview REST Basics (REST Architecture, REST Principles, Client-Server Model, Statelessness, Resources & URIs) HTTP Methods(GET, POST, PUT, PATCH, DELETE) Headers, Object Mapper Class JSON Overview, XML vs JSON, Jackson API, GSON API Creation of Rest API, Converting Java to JSON and Vice Versa using GSON API REST API Annotations(@RequestMapping, @RequestBody, @ResponseBody, @RequestParam, @PathVariable) Request & Response Structure Media Type, Producer, Consumer Path Variables, Query Parameters Run Time Operations (Marshalling, Un Marshalling) ResponseEntity Class Custom Responses, Error Responses

		Status Codes, PostMan, Swagger & Swagger UI Interpropability Rest template
6	Spring Security	Introduction to Spring Security Spring Security Architecture Security Filter Chain Authentication vs Authorization Authentication Providers User Details Service Password Encoding Form Login & Custom Login CSRF Protection Session Management CORS (Cross-Origin Resource Sharing) JWT (JSON Web Token) Authentication OAuth2 Method-Level Security (@EnableMethodSecurity, @PreAuthorize)
7	Spring Scheduling	Scheduling overview @EnableScheduling @Scheduled fixedRate fixedDelay Corn-Expression Practice using example
8	Microservices	Monolith Architecture Introduction Monolith Application Deployment Process Load balancer (Cluster) Load Balancing Algorithms a) Round Robin Monolith Architecture Drawbacks Micro services Introduction Micro Services Advantages Micro Services Dis-Advantages Micro services Architecture Micro services Development Service Communication's (Rest Template, Web Client, Feign Client) Interservice communication (FeignClient) Service Registry (Netflix Eureka) API Gateway Resilience4j (Circuit Breaker) Spring Boot Admin Server Spring Boot Admin Client Distributed Logging a) Zipkin Server ConfigServer Introduction & Implementation, Saga Pattern
9	Spring Boot Testing (JUnit, Mockito)	JUnit Annotations & Test Lifecycle Assertions in JUnit Mockito Annotations Stubbing in Mockito Verification in Mockito @SpringBootTest @WebMvcTest + MockMvc @MockBean vs @Mock Layer-wise Testing Exception Testing
10	Tools	Maven Gradle Lombok Git & GitHub Log4j & Slf4j JIRA Docker

1. Spring Core

Interview Questions

1. Fundamentals of Spring Core

1. What is Spring Framework?
2. Why do we use Spring?
3. What problems does Spring solve?
4. What are the main features of Spring?
5. Difference between Spring and Spring Boot?
6. What are modules in Spring Framework?
7. What is Spring Context?
8. What is Spring container?
9. Role of Spring container?
10. Why is Spring called lightweight?
11. Advantages of Spring over traditional Java development?
12. What is POJO in Spring?
13. What is Dependency Management in Spring?

2. Inversion of Control (IoC)

14. What is IoC?
15. Why is IoC needed?
16. What is IoC container?
17. How does Spring implement IoC?
18. Difference between traditional object creation and IoC?
19. What is loose coupling?
20. How IoC supports loose coupling?
21. Advantages of IoC?
22. Real-time example of IoC?
23. Difference between IoC and Factory Pattern?

3. Dependency Injection (DI)

- 24. What is Dependency Injection?
- 25. Why is DI needed?
- 26. How DI is related to IoC?
- 27. Types of Dependency Injection?
- 28. Constructor Injection?
- 29. Setter Injection?
- 30. Field Injection?
- 31. Which injection is preferred and why?

@Autowired Modes

- 32. What is @Autowired?
- 33. What is autowiring by type?
- 34. What is autowiring by name?
- 35. What is autowiring by constructor?
- 36. What is @Qualifier?
- 37. What is @Primary?
- 38. What happens when multiple beans of same type exist?
- 39. What is circular dependency?
- 40. How Spring resolves dependencies?
- 41. How to inject primitive values?

4. Spring Beans

- 42. What is a Spring bean?
- 43. Difference between object and Spring bean?
- 44. How is a bean created?
- 45. What is bean definition?
- 46. What is bean wiring?
- 47. What is component scanning?
- 48. What is @Component?
- 49. Difference between @Component, @Service, @Repository?
- 50. What is @Bean?
- 51. Difference between @Bean and @Component?

5. Bean Lifecycle

- 52. What is bean lifecycle?
- 53. What are the phases of bean lifecycle?
- 54. What happens during bean instantiation?
- 55. What is dependency injection phase?
- 56. What is initialization phase?
- 57. What is destroy phase?
- 58. What is initialization method?
- 59. What is destroy method?
- 60. What is @PostConstruct?
- 61. What is @PreDestroy?
- 62. Constructor vs init method?
- 63. What is BeanPostProcessor?
- 64. How BeanPostProcessor works?

6. Bean Scopes

- 65. What is bean scope?
- 66. What are bean scopes in Spring?

Core Scopes

- 67. What is Singleton scope (default)?
- 68. What is Prototype scope?

Web Scopes

- 69. What is Request scope?
- 70. What is Session scope?
- 71. What is Global Session scope?

Advanced

- 72. Why singleton is default scope?
- 73. Difference between singleton and prototype?
- 74. Which scope is used for shopping cart?
- 75. Can singleton use prototype bean?

7. BeanFactory vs ApplicationContext

- 76. What is BeanFactory?

- 77. What is ApplicationContext?
- 78. Difference between BeanFactory and ApplicationContext?
- 79. What is lazy loading?
- 80. What is eager loading?
- 81. Why ApplicationContext is preferred?
- 82. Types of ApplicationContext?

8. Bean Configuration

- 83. What is bean configuration?
- 84. Ways to configure beans?

XML Configuration

- 85. What is XML configuration?
- 86. Advantages and disadvantages of XML config?

Java-Based Configuration

- 87. What is Java-based configuration?
- 88. What is @Configuration?
- 89. Why Java config preferred over XML?

Annotation-Based Configuration

- 90. What is annotation-based configuration?
- 91. What is @ComponentScan?
- 92. How annotation scanning works?

9. Environment & Properties

- 93. What is application.properties?
- 94. Why externalize configuration?
- 95. What is @Value annotation?
- 96. How @Value works internally?
- 97. Difference between .properties and .yaml?
- 98. What is @ConfigurationProperties?
- 99. What is Spring profile?
- 100. How to activate profiles?
- 101. How to manage different environments (dev/test/prod)?

10. Scenario-Based Questions

- 102. Why constructor injection is preferred over field injection?
- 103. If two beans of same type exist what happens?
- 104. How does Spring create singleton bean?
- 105. What happens if dependency bean is missing?
- 106. How do you switch database configuration between environments?
- 107. Explain complete bean lifecycle with example.

11. Annotation Quick Questions

- 108. What is @Component?
- 109. What is @Autowired?
- 110. What is @Bean?
- 111. What is @Configuration?
- 112. What is @Qualifier?
- 113. What is @Primary?
- 114. What is @Value?
- 115. What is @PostConstruct?
- 116. What is @PreDestroy?

2. Spring Boot Core

Interview Questions

1. Introduction to Spring Boot

1. What is Spring Boot?
2. Why was Spring Boot introduced?
3. Difference between Spring and Spring Boot?
4. What are features of Spring Boot?
5. Why is Spring Boot called opinionated framework?
6. What is convention over configuration?
7. What is standalone application?
8. Explain Spring Boot architecture.

2. Spring Boot Project Setup

9. What is Spring Initializer?
10. How to create Spring Boot project?
11. What is spring-boot-starter-parent?
12. Difference between Maven and Gradle?
13. Explain standard project structure.

3. Spring Boot Starters

14. What are starter dependencies?
15. What is spring-boot-starter-web?
16. What is spring-boot-starter-data-jpa?
17. How starters simplify dependency management?

4. Auto Configuration

18. What is auto configuration?
19. How does auto configuration work?
20. What is @EnableAutoConfiguration?
21. How to disable auto configuration?

5. Spring Boot Internals

- 22. What is `@SpringBootApplication`?
- 23. What are components of `@SpringBootApplication`?
- 24. How does `@SpringBootApplication` work internally?
- 25. What happens inside `SpringApplication.run()`?
- 26. What is `SpringApplication` class?
- 27. What is `ApplicationContext` in Spring Boot?

6. Application Properties & Configuration

- 28. What is `application.properties`?
- 29. Difference between `.properties` and `.yaml`?
- 30. How to configure datasource properties?
- 31. What is `@Value`?
- 32. What is `@ConfigurationProperties`?
- 33. How configuration binding works?

7. Profiles

- 34. What are profiles?
- 35. Why use profiles?
- 36. How to activate profiles?
- 37. Difference between dev/test/prod profiles?
- 38. Can multiple profiles be active?
- 39. What is `@Profile` annotation?

8. Embedded Servers

- 40. What is embedded server?
- 41. Default embedded server in Spring Boot?
- 42. Difference between embedded and external server?
- 43. How to change Tomcat to Jetty or Undertow?
- 44. How to change server port?

9. Spring Boot DevTools

- 45. What is DevTools?
- 46. What is automatic restart?
- 47. What is LiveReload?
- 48. What are advantages of DevTools?

10. Spring Boot Logging

- 49. What is default logging framework?
- 50. Explain logging levels.
- 51. Difference between Logger and System.out.println?
- 52. What is @Slf4j?
- 53. How to customize logging configuration?

11. Spring Boot Actuator

- 54. What is Actuator?
- 55. Common actuator endpoints?
- 56. What is /health endpoint?
- 57. What is /info endpoint?
- 58. How to expose actuator endpoints?
- 59. How to secure actuator endpoints?

12. Exception Handling

- 60. What is exception handling in Spring Boot?
- 61. What is @ExceptionHandler?
- 62. What is @ControllerAdvice?
- 63. What is @RestControllerAdvice?
- 64. How to handle global exceptions?
- 65. Difference between checked and unchecked exceptions?

13. Externalized Configuration

- 66. What is externalized configuration?
- 67. Why external configuration is important?
- 68. What are environment variables?

- 69. How command-line arguments work?
- 70. What is property precedence order?
- 71. How Spring Boot resolves properties?

14. Runners

- 72. What is CommandLineRunner?
- 73. What is ApplicationRunner?
- 74. Difference between CommandLineRunner and ApplicationRunner?
- 75. When should you use runners?

15. Scenario-Based Questions

- 76. How auto configuration works internally?
- 77. How does @SpringBootApplication work?
- 78. How do you handle global exceptions?
- 79. How to change embedded server?
- 80. How to manage different environments?

16. Annotation Quick Questions

- 81. What is @SpringBootApplication?
- 82. What is @EnableAutoConfiguration?
- 83. What is @ConfigurationProperties?
- 84. What is @Value?
- 85. What is @RestController?
- 86. What is @RequestMapping?
- 87. What is @GetMapping?

3. Spring Data JPA

Interview Questions

1. JDBC Template

1. What is JdbcTemplate?
2. Why JdbcTemplate is used?
3. What problems does it solve compared to JDBC?
4. What are main methods in JdbcTemplate?
5. Difference between query(), queryForObject(), update()?
6. What is RowMapper?
7. What is ResultSetExtractor?
8. Difference between RowMapper and ResultSetExtractor?
9. What is BeanPropertyRowMapper?
10. How JdbcTemplate handles exceptions?

2. JPA & ORM Basics

11. What is JPA?
12. What is ORM?
13. What problems does JPA solve?
14. Advantages of JPA?
15. What is EntityManager?
16. What is persistence context?
17. What are entity lifecycle states?
18. What is dirty checking?
19. What is first-level cache?
20. What is flushing?
21. What is write-behind mechanism?
22. How transaction management works in JPA?

3. Hibernate Overview

23. What is Hibernate?

- 24. Difference between Hibernate and JPA?
- 25. What is Hibernate architecture?
- 26. What is Session vs EntityManager?
- 27. What is second-level cache?

4. Spring Data JPA Overview

- 28. What is Spring Data JPA?
- 29. How Spring Data JPA works internally?
- 30. What is repository pattern?
- 31. What is repository proxy?

5. Repository Layer

- 32. What is CrudRepository?
- 33. What methods are available in CrudRepository?
- 34. What is JpaRepository?
- 35. Difference between CrudRepository and JpaRepository?
- 36. What is PagingAndSortingRepository?

6. Database Operations

- 37. What does save() do?
- 38. Difference between save() and saveAll()?
- 39. What is findAll()?
- 40. What is findById()?
- 41. What is delete()?
- 42. How bulk operations work?

7. HikariCP Connection Pool

- 43. What is connection pooling?
- 44. What is HikariCP?
- 45. Why HikariCP is default in Spring Boot?
- 46. What is max pool size?
- 47. What is idle timeout?
- 48. What is connection timeout?

- 49. What is pool exhaustion?
- 50. How to tune HikariCP?

8. Entities, DTOs & DAO

- 51. What is an Entity class?
- 52. What annotations are used in Entity?
- 53. What is @Entity?
- 54. What is @Id?
- 55. What is @GeneratedValue?
- 56. What is GenerationType?
- 57. What is @Column?
- 58. What is @Table?
- 59. What is DTO?
- 60. Difference between Entity and DTO?
- 61. What is DAO layer?

9. Derived Query Methods (findByXXX)

- 62. What are findBy methods?
- 63. How method naming works?
- 64. Rules for writing findBy methods?
- 65. Limitations of derived queries?
- 66. What is query creation strategy?

10. Query By Example (QBE)

- 67. What is Query By Example?
- 68. Why use QBE?
- 69. Advantages and limitations of QBE?

11. JPQL & Native Queries

- 70. What is JPQL?
- 71. What is native query?
- 72. Difference between JPQL and SQL?
- 73. What is @Query annotation?

- 74. What are named parameters?
- 75. What are positional parameters?
- 76. How to map results to DTO?
- 77. What is result set mapping?

12. Pagination & Sorting

- 78. What is pagination?
- 79. What is Pageable?
- 80. What is Page?
- 81. What is Sort?
- 82. How pagination improves performance?
- 83. How to combine pagination and sorting?

13. Embedded Database (H2)

- 84. What is H2 database?
- 85. What is in-memory database?
- 86. How to enable H2 console?
- 87. Difference between H2 and MySQL?
- 88. What is schema.sql?
- 89. How to preload data?

14. Auditing

- 90. What is auditing?
- 91. What is @CreatedDate?
- 92. What is @LastModifiedDate?
- 93. How to enable auditing?

15. Relationships

- 94. What is OneToOne?
- 95. What is OneToMany?
- 96. What is ManyToOne?
- 97. What is ManyToMany?
- 98. What is mappedBy?

99. What is fetch type?

100. What is cascade?

16. Transactions

101. What is @Transactional?

102. How transaction management works?

103. What is propagation?

104. What is isolation level?

17. Performance Issues

105. What is N+1 problem?

106. Why N+1 problem occurs?

107. How to solve N+1 using JOIN FETCH?

108. How to solve N+1 using @EntityGraph?

18. Scenario-Based Questions

109. Entity updated without save(), why?

110. Same query hits DB once, why?

111. Tables deleted on restart, why?

112. DB connection issues, what to check?

113. save() called twice, what happens?

114. Data not visible immediately, why?

115. API slow with large data, solution?

116. How to write custom query when method name fails?

117. Too many queries executing, why?

118. DB connections exhausted, why?

4. Spring MVC

Interview Questions

1. Introduction to Spring MVC

1. What is Spring MVC?
2. What does MVC stand for?
3. What is Model-View-Controller pattern?
4. What problem does MVC solve?
5. Features of Spring MVC?
6. Difference between Spring and Spring MVC?

2. Spring MVC Architecture

7. Explain Spring MVC architecture.
8. What is Model in MVC?
9. What is View in MVC?
10. What is Controller in MVC?
11. How do Model, View, Controller interact?
12. What is Front Controller pattern?
13. What is DispatcherServlet?
14. Why DispatcherServlet is called Front Controller?

3. Spring MVC Request Flow

15. Explain Spring MVC request flow.
16. What are steps in request lifecycle?
17. What is HandlerMapping?
18. What is HandlerAdapter?
19. What is ViewResolver?
20. How request reaches controller?
21. How response returns to client?
22. End-to-end flow: Client → DispatcherServlet → HandlerMapping → Controller → Service → DAO → ViewResolver → View

4. Spring MVC Configuration & Embedded Servers

- 23. How to configure Spring MVC?
- 24. What is Java-based configuration in MVC?
- 25. What is @EnableWebMvc?
- 26. What is WebMvcConfigurer?
- 27. What are embedded servers?
- 28. Difference between Tomcat, Jetty, Undertow?
- 29. How to change embedded server?
- 30. How Spring Boot integrates with Spring MVC?

5. Controller Layer

- 31. What is @Controller?
- 32. What is @RestController?
- 33. Difference between @Controller and @RestController?
- 34. What is @RequestMapping?
- 35. Difference between @GetMapping, @PostMapping, @PutMapping, @DeleteMapping?
- 36. What is @ResponseBody?
- 37. What is @RequestBody?

Request Parameters

- 38. What is @RequestParam (Query Param)?
- 39. What is @PathVariable (Path Param)?
- 40. Difference between QueryParam and PathParam?

6. DTO, DAO & Layered Architecture

- 41. What is DTO?
- 42. What is DAO?
- 43. Difference between Entity and DTO?
- 44. Difference between DTO and DAO?
- 45. Why DTO is used?
- 46. Why DAO layer is used?
- 47. What is Mapper?
- 48. Explain layered architecture (Controller → Service → Repository)?

7. Validation

- 49. What is validation in Spring MVC?
- 50. Why validation is needed?
- 51. What is @Valid?
- 52. What is @Validated?
- 53. What is BindingResult?
- 54. What are common validation annotations (@NotNull, @NotBlank, @Size, @Email)?
- 55. How validation works internally?
- 56. How to validate request DTO?

8. Session Management

- 57. What is session management?
- 58. What is HTTP Session?
- 59. How Spring MVC manages session?
- 60. What is HttpSession?
- 61. What is @SessionAttributes?
- 62. Difference between session and cookies?
- 63. What is session timeout?
- 64. Use cases of session management?

9. View Technologies (Thymeleaf)

- 65. What is Thymeleaf?
- 66. Why use Thymeleaf in Spring MVC?
- 67. How Thymeleaf integrates with Spring MVC?
- 68. What is template engine?
- 69. Difference between JSP and Thymeleaf?

10. Scenario-Based Questions

- 70. Explain login flow using Spring MVC.
- 71. How request flows from browser to database and back?
- 72. What happens if URL not mapped?
- 73. How validation works in form submission?
- 74. Role of service layer?

11. Coding Questions

- 75. Write simple controller example.
- 76. Write validation example using `@Valid`.
- 77. Write path variable mapping example.
- 78. Design layered MVC flow for user registration.
- 79. Create session management example.

5. REST API & Web Services

Interview Questions

1. Distributed Applications & API Basics

1. What is a Distributed Application?
2. What is an API?
3. What is Web API?
4. Difference between API and Web Service?
5. What is an endpoint?
6. What is request and response?

2. SOAP Overview

7. What is SOAP?
8. What is WSDL?
9. When should SOAP be used?
10. Advantages and disadvantages of SOAP?
11. Difference between SOAP and REST?

3. REST Overview

12. What is REST?
13. What is RESTful API?
14. Difference between REST and SOAP?
15. Advantages of REST?

4. REST Basics

16. What is REST architecture?
17. What are REST principles?
18. What is statelessness?
19. What is client-server model?
20. What is a resource?
21. What is URI vs URL?

5. HTTP Methods

- 22. What is GET?
- 23. What is POST?
- 24. What is PUT?
- 25. What is PATCH?
- 26. What is DELETE?
- 27. Difference between PUT and PATCH?
- 28. What are safe and idempotent methods?

6. HTTP Headers & Object Mapping

- 29. What are HTTP headers?
- 30. What is Content-Type?
- 31. What is Accept header?
- 32. What is ObjectMapper class?
- 33. How ObjectMapper converts JSON to Java?

7. JSON & Data Formats

- 34. What is JSON?
- 35. What is XML?
- 36. Difference between JSON and XML?
- 37. What is Jackson API?
- 38. What is Gson API?
- 39. Difference between Jackson and Gson?

8. REST API Creation (Spring Boot)

- 40. How to create REST API?
- 41. What is @RestController?
- 42. What is @RequestMapping?
- 43. What is @GetMapping?
- 44. What is @PostMapping?
- 45. What is @PutMapping?
- 46. What is @PatchMapping?
- 47. What is @DeleteMapping?

9. Request & Response Handling

- 48. What is request structure?
- 49. What is response structure?
- 50. What is @RequestBody?
- 51. What is @ResponseBody?
- 52. What is @RequestParam?
- 53. What is @PathVariable?
- 54. Difference between QueryParam and PathParam?

10. Media Types & Content Negotiation

- 55. What is media type?
- 56. What is producer and consumer?
- 57. What is application/json?
- 58. What is application/xml?
- 59. What is content negotiation?

11. Runtime Processing

- 60. What is marshallng?
- 61. What is unmarshalling?
- 62. How Java objects converted to JSON?
- 63. How JSON converted to Java objects?

12. ResponseEntity & Error Handling

- 64. What is ResponseEntity?
- 65. Why use ResponseEntity?
- 66. What is custom response?
- 67. What is error response?
- 68. How to handle global exceptions?
- 69. What is @ExceptionHandler?
- 70. What is @ControllerAdvice?
- 71. What is @RestControllerAdvice?

13. HTTP Status Codes

- 72. What are HTTP status codes?
- 73. What are 1xx codes?
- 74. What are 2xx codes?
- 75. What are 3xx codes?
- 76. What are 4xx codes?
- 77. What are 5xx codes?
- 78. Difference between 401 and 403?
- 79. Important status codes (200, 201, 204, 400, 401, 403, 404, 500)?

14. API Testing & Documentation

- 80. What is Postman?
- 81. How to test APIs using Postman?
- 82. What is Swagger?
- 83. What is Swagger UI?
- 84. Benefits of API documentation?

15. Interoperability

- 85. What is interoperability?
- 86. Why interoperability is important?

16. RestTemplate

- 87. What is RestTemplate?
- 88. How to call GET API using RestTemplate?
- 89. How to call POST API using RestTemplate?
- 90. Difference between RestTemplate and WebClient?

17. Advanced API Concepts

- 91. What is idempotency?
- 92. Difference between safe and idempotent methods?
- 93. What is pagination in APIs?
- 94. What is API versioning?
- 95. What is authentication in APIs?
- 96. Difference between Basic Auth and JWT?

18. Coding & Scenario-Based Questions

- 97. Create GET endpoint example.
- 98. Create POST endpoint example.
- 99. Use PathVariable example.
- 100. Use QueryParam example.
- 101. Return custom response using ResponseEntity.
- 102. How to handle errors globally?
- 103. How to validate user input?
- 104. How to call third-party APIs?

6.Spring Security

Interview Questions

1. Introduction to Spring Security

1. What is Spring Security?
2. Why do we use Spring Security?
3. What problems does it solve?
4. What happens when spring-boot-starter-security is added?
5. What are default credentials in Spring Boot?

2. Spring Security Architecture

6. Explain Spring Security architecture.
7. What are core components of Spring Security?
8. What is SecurityContext?
9. What is SecurityContextHolder?
10. What is Authentication object?
11. What is GrantedAuthority?
12. How request flows internally in Spring Security?

3. Security Filter Chain

13. What is SecurityFilterChain?
14. Why filter chain is required?
15. What is filter ordering?
16. What is OncePerRequestFilter?
17. What is UsernamePasswordAuthenticationFilter?
18. How to add custom filter?
19. Where to place JWT filter in chain?

4. Security Configuration

20. How to configure Spring Security using SecurityFilterChain bean?
21. What is HttpSecurity?

- 22. What is `authorizeHttpRequests()`?
- 23. What is `requestMatchers()`?
- 24. What is `permitAll()` vs `authenticated()`?
- 25. How to disable CSRF?
- 26. How to make application stateless?

5. Authentication vs Authorization

- 27. What is authentication?
- 28. What is authorization?
- 29. Difference with example?
- 30. How Spring handles both?

6. Roles & Authorities

- 31. What are roles?
- 32. What are authorities?
- 33. Difference between roles and authorities?
- 34. What is `hasRole()`?
- 35. What is `hasAuthority()`?

7. Authentication Providers

- 36. What is `AuthenticationProvider`?
- 37. What is `AuthenticationManager`?
- 38. What is `DaoAuthenticationProvider`?
- 39. Can we have multiple providers?
- 40. How to implement custom authentication?

8. UserDetailsService

- 41. What is `UserDetailsService`?
- 42. What is `loadUserByUsername()`?
- 43. What happens if user not found?
- 44. What is `UserDetails`?
- 45. How to connect with database?

9. Password Encoding

- 46. Why password encoding required?
- 47. What is PasswordEncoder?
- 48. What is BCryptPasswordEncoder?
- 49. Why BCrypt is slow?
- 50. What is salting?

10. Form Login & Custom Login

- 51. What is form-based authentication?
- 52. How to configure custom login page?
- 53. What is login processing URL?
- 54. What are success and failure handlers?

11. CSRF Protection

- 55. What is CSRF?
- 56. How CSRF attack works?
- 57. Why enabled by default?
- 58. When to disable CSRF?
- 59. What is CSRF token?

12. Session Management

- 60. What is session-based authentication?
- 61. What is stateless authentication?
- 62. How to configure session?
- 63. How to make stateless app?

13. CORS

- 64. What is CORS?
- 65. Why CORS error occurs?
- 66. How to enable CORS?
- 67. @CrossOrigin vs global config?

14. JWT Authentication

- 68. What is JWT?
- 69. Structure of JWT?
- 70. What is header, payload, signature?
- 71. How JWT is generated?
- 72. How JWT is validated?
- 73. What is secret key?
- 74. What is expiration?
- 75. What is access token?
- 76. What is refresh token?
- 77. JWT vs session?
- 78. JWT authentication flow step-by-step?

15. Exception Handling in Security

- 79. What is AuthenticationEntryPoint?
- 80. What is AccessDeniedHandler?
- 81. Difference between them?
- 82. How to handle unauthorized access?

16. OAuth2

- 83. What is OAuth2?
- 84. What are OAuth2 flows?
- 85. What is authorization code flow?
- 86. What is client credentials flow?
- 87. What is resource server?
- 88. What is authorization server?
- 89. OAuth2 vs JWT?
- 90. What is OpenID Connect?

17. Method-Level Security

- 91. What is method-level security?
- 92. What is @EnableMethodSecurity?
- 93. What is @PreAuthorize?
- 94. What is @Secured?

95. How to restrict method access?

18. Testing Spring Security

96. How to test secured APIs?

97. What is @WithMockUser?

98. How to test using MockMvc?

99. How to test JWT endpoints?

19. Scenario-Based Questions

100. How did you implement security in your project?

101. How to secure REST APIs?

102. How to implement JWT authentication?

103. How to restrict endpoints by role?

104. How to test secured APIs in Postman?

105. What challenges did you face?

7. Spring Scheduling

Interview Questions

1. Scheduling Fundamentals

1. What is scheduling in Spring?
2. Why do we use scheduling?
3. What is task scheduling?
4. What are real-time use cases of scheduling?
5. What is a scheduler?
6. Difference between scheduling and asynchronous processing?

2. Spring Scheduling Architecture

7. How scheduling works internally in Spring?
8. What is TaskScheduler?
9. What is ThreadPoolTaskScheduler?
10. How threads are managed in scheduled tasks?

3. @EnableScheduling

11. What is @EnableScheduling?
12. Why is it required?
13. Where should it be placed?
14. What happens if it is missing?
15. How Spring enables scheduling internally?

4. @Scheduled Annotation

16. What is @Scheduled?
17. How does @Scheduled work?
18. What parameters are available in @Scheduled?
19. What are rules for @Scheduled methods?
20. Can scheduled methods take parameters?

5. fixedRate

- 21. What is fixedRate?
- 22. How fixedRate works?
- 23. When should fixedRate be used?
- 24. What happens if execution exceeds fixedRate?

6. fixedDelay

- 25. What is fixedDelay?
- 26. How fixedDelay works?
- 27. When should fixedDelay be used?
- 28. Difference between fixedDelay and fixedRate?

7. Cron Expressions

- 29. What is cron expression?
- 30. Structure of cron expression?
- 31. Meaning of each field?
- 32. Special characters (*, ?, -, /, ,)?
- 33. Difference between cron and fixedRate?
- 34. Cron for every 5 seconds?
- 35. Cron for every minute?
- 36. Cron for midnight execution?

8. Threading & Parallel Execution

- 37. Can multiple scheduled tasks run in parallel?
- 38. How to enable parallel execution?
- 39. What is thread pool in scheduling?
- 40. How to configure ThreadPoolTaskScheduler?

9. Exception Handling in Scheduling

- 41. What happens if scheduled task throws exception?
- 42. How to handle exceptions in scheduled jobs?
- 43. How to ensure task retry?

10. Preventing Overlapping Tasks

- 44. What is overlapping in scheduled jobs?
- 45. How to prevent overlapping execution?
- 46. What is synchronized scheduling?
- 47. How to use locks in scheduling?

11. Advanced Scheduling Concepts

- 48. What is dynamic scheduling?
- 49. How to change cron expression at runtime?
- 50. What is scheduling with database triggers?
- 51. What is distributed scheduling?
- 52. What is Quartz Scheduler?
- 53. Difference between Spring Scheduler and Quartz?

12. Real-World Use Cases

- 54. How to schedule database cleanup job?
- 55. How to send daily notification emails?
- 56. How to schedule report generation?
- 57. How to run background batch jobs?

13. Scenario-Based Questions

- 58. What happens if application restarts during task?
- 59. How to ensure job runs only once in distributed system?
- 60. How to monitor scheduled tasks?
- 61. How to log execution of jobs?

14. Coding Questions

- 62. Write fixedRate example.
- 63. Write fixedDelay example.
- 64. Write cron expression example.
- 65. Configure thread pool scheduler.
- 66. Prevent overlapping example.

8. Microservices

Interview Questions

1. Monolith Architecture

Introduction

1. What is Monolithic Architecture?
2. What are the characteristics of monolithic applications?
3. How does a monolithic system differ from distributed systems?

Deployment Process

4. How is a monolithic application deployed?
5. What are the challenges in monolith deployment?

Load Balancer & Cluster

6. What is a Load Balancer?
7. What is clustering in monolithic systems?

Load Balancing Algorithms

8. What is Round Robin algorithm?
9. What are other load balancing algorithms?

Drawbacks

10. What are the limitations of monolithic architecture?
11. Why monoliths become hard to scale?

2. Introduction to Microservices

12. What are Microservices?
13. Difference between Monolith and Microservices?
14. When should you use Microservices?
15. What are the key characteristics of Microservices?

Advantages

16. What are advantages of Microservices?

Disadvantages

17. What are disadvantages of Microservices?

18. What challenges are faced in Microservices?

3. Microservices Architecture

19. What is Microservices Architecture?

20. What are core principles of Microservices design?

21. What is loosely coupled service?

22. What is bounded context?

23. What is database-per-service pattern?

4. Microservices Development

24. What is Microservices development lifecycle?

25. How do you design a Microservice?

26. What tools are used in Microservices development?

5. Service Communication

27. What is synchronous communication?

28. What is asynchronous communication?

29. What is RestTemplate?

30. What is WebClient?

31. RestTemplate vs WebClient?

32. What is Feign Client?

33. Feign vs RestTemplate?

34. How does one service call another?

Interservice Communication

35. How does FeignClient work internally?

6. Service Registry (Eureka)

36. What is Service Registry?

37. What is Eureka Server?

38. What is Eureka Client?

39. How do services register with Eureka?

40. What is self-preservation mode in Eureka?

7. Service Discovery

41. What is service discovery?

42. Why is service discovery needed?

43. Client-side vs server-side discovery?

44. How does Eureka support discovery?

8. API Gateway

- 45. What is API Gateway?
- 46. Why is API Gateway needed?
- 47. How does API Gateway work?
- 48. What is routing in API Gateway?
- 49. What is Spring Cloud Gateway?

9. Load Balancing

- 50. What is load balancing?
- 51. Client-side vs server-side load balancing?
- 52. What is Spring Cloud LoadBalancer?
- 53. How load balancing works with Eureka?

10. Circuit Breaker (Resilience4j)

- 54. What is Circuit Breaker pattern?
- 55. What is Resilience4j?
- 56. What are circuit breaker states?
- 57. Explain closed, open, half-open states.
- 58. What is fallback method?
- 59. Difference between retry and circuit breaker?

11. Spring Boot Admin

- 60. What is Spring Boot Admin Server?
- 61. Why use Admin Server?
- 62. Difference between Actuator and Admin Server?
- 63. What is Spring Boot Admin Client?
- 64. How does Admin Client register with server?

12. Distributed Logging

- 65. What is distributed logging?
- 66. Why is logging important in Microservices?
- 67. What is Zipkin?
- 68. How Zipkin works?
- 69. What is trace and span in Zipkin?

13. Configuration Management

- 70. What is centralized configuration?
- 71. What is Config Server?

- 72. What is Config Client?
- 73. Difference between local config and centralized config?

Implementation

- 74. How to implement Config Server?
- 75. How services read configuration from Config Server?

14. Saga Pattern

- 76. What is Saga Pattern?
- 77. Why Saga is used in Microservices?
- 78. Types of Saga (Choreography vs Orchestration)?
- 79. How Saga handles distributed transactions?

15. Scenario-Based Questions

- 80. Explain Microservice request flow using API Gateway and Eureka.
- 81. How service A calls service B using Feign Client?
- 82. What happens if one service is down?
- 83. How to make communication fault-tolerant?

16. Annotation Questions

- 84. What is `@EnableEurekaServer`?
- 85. What is `@EnableDiscoveryClient`?
- 86. What is `@FeignClient`?
- 87. What is `@LoadBalanced`?
- 88. What is `@CircuitBreaker`?

17. Coding Questions

- 89. Steps to create Eureka Server.
- 90. Write Feign Client example.
- 91. Write API Gateway routing example.
- 92. Write Circuit Breaker fallback example.

9. Spring Testing (JUnit & Mockito)

Interview Questions

1. Testing Fundamentals

1. What is unit testing?
2. What is integration testing?
3. Difference between unit testing and integration testing?
4. Why testing is important?
5. What is TDD (Test Driven Development)?

2. JUnit Basics

6. What is JUnit?
7. What is JUnit 5?
8. Difference between JUnit 4 and JUnit 5?

3. JUnit Annotations & Test Lifecycle

9. What are JUnit annotations?
10. What is @Test?
11. What is @BeforeEach?
12. What is @AfterEach?
13. What is @BeforeAll?
14. What is @AfterAll?
15. What is @DisplayName?
16. What is @Disabled?
17. What is test lifecycle in JUnit?
18. Order of execution of JUnit lifecycle methods?

4. Assertions in JUnit

19. What are assertions?
20. What is assertEquals()?
21. What is assertNotEquals()?

- 22. What is assertTrue()?
- 23. What is assertFalse()?
- 24. What is assertNull()?
- 25. What is assertNotNull()?
- 26. What is assertThrows()?
- 27. What is assertAll()?
- 28. Difference between assertEquals and assertEquals?

5. Mockito Basics

- 29. What is Mockito?
- 30. Why Mockito is used?
- 31. What is mocking?
- 32. What is spy?
- 33. Difference between mock and spy?

6. Mockito Annotations

- 34. What is @Mock?
- 35. What is @InjectMocks?
- 36. What is @Spy?
- 37. What is @Captor?
- 38. What is @MockBean?
- 39. What is @ExtendWith(MockitoExtension.class)?
- 40. Difference between @Mock and @MockBean?

7. Stubbing in Mockito

- 41. What is stubbing?
- 42. What is when().thenReturn()?
- 43. What is thenThrow()?
- 44. What is doReturn()?
- 45. Difference between when() and doReturn()?
- 46. What is BDDMockito?
- 47. What is given().willReturn()?

8. Verification in Mockito

- 48. What is verification?
- 49. What is verify()?
- 50. How to verify method call count?
- 51. What is verifyNoInteractions()?
- 52. What is verifyNoMoreInteractions()?
- 53. What is times(), atLeast(), atMost()?
- 54. What is ArgumentCaptor?

9. Spring Boot Testing Annotations

- 55. What is @SpringBootTest?
- 56. When to use @SpringBootTest?
- 57. What is @WebMvcTest?
- 58. What is MockMvc?
- 59. How to test controller using MockMvc?
- 60. Difference between @SpringBootTest and @WebMvcTest?

10. Layer-wise Testing

- 61. What is layer-wise testing?
- 62. How to test controller layer?
- 63. How to test service layer?
- 64. How to test repository layer?
- 65. What is @DataJpaTest?
- 66. When to use slice testing?

11. Exception Testing

- 67. How to test exceptions in JUnit?
- 68. What is assertThrows()?
- 69. How to test custom exceptions?
- 70. How to test global exception handler?

12. Advanced Testing Concepts

- 71. What is test coverage?

- 72. What is code coverage tools (JaCoCo)?
- 73. What is parameterized test?
- 74. What is @ParameterizedTest?
- 75. What is @ValueSource?
- 76. What is @CsvSource?
- 77. What is test container?
- 78. What is integration testing with database?

13. Scenario-Based Questions

- 79. How do you test service layer with dependencies?
- 80. How do you mock repository in service test?
- 81. How to test REST API endpoints?
- 82. How to handle null values in test?
- 83. How to test validation errors?
- 84. How to test exception flow in controller?
- 85. What happens if dependency is not mocked?

14. Coding Questions

- 86. Write unit test for service class.
- 87. Write test using Mockito for repository call.
- 88. Write controller test using MockMvc.
- 89. Write test for exception handling.
- 90. Write parameterized test example.

10. Tools & DevOps

Interview Questions

1. Maven

1. What is Maven?
2. What is build automation?
3. What is POM.xml?
4. What are Maven dependencies?
5. What is Maven lifecycle?
6. Phases in Maven lifecycle?
7. What is clean, install, package?
8. What are plugins in Maven?
9. What is dependency scope?
10. What is dependency management?
11. How Maven resolves dependencies?
12. What is Maven repository (local, central, remote)?
13. What is transitive dependency?
14. How to exclude dependency?
15. Difference between compile and runtime scope?

2. Gradle

16. What is Gradle?
17. Difference between Maven and Gradle?
18. What is build.gradle file?
19. What is Gradle DSL?
20. What are tasks in Gradle?
21. What is Gradle wrapper?
22. How dependency management works in Gradle?
23. Advantages of Gradle over Maven?

3. Lombok

24. What is Lombok?
25. Why Lombok is used?
26. What problem does Lombok solve?
27. What is @Getter and @Setter?
28. What is @Data?
29. What is @Builder?
30. What is @AllArgsConstructor?
31. What is @NoArgsConstructor?
32. What is @RequiredArgsConstructor?
33. What is @ToString?
34. What is @EqualsAndHashCode?
35. What are disadvantages of Lombok?

4. Git & GitHub

36. What is Git?
37. What is version control system?
38. Difference between Git and GitHub?
39. What is repository?
40. What is commit?
41. What is branch?
42. What is merge?
43. What is rebase?
44. What is pull request?
45. What is clone, pull, push?
46. What is staging area?
47. What is git stash?
48. What is git reset vs revert?
49. What is merge conflict?
50. How to resolve merge conflicts?
51. What is .gitignore?
52. What is fork?

5. Logging (Log4j & SLF4J)

- 53. What is logging?
- 54. Why logging is important?
- 55. What is Log4j?
- 56. What is SLF4J?
- 57. Difference between Log4j and SLF4J?
- 58. What is logging level?
- 59. What are log levels (INFO, DEBUG, ERROR, WARN)?
- 60. What is logger configuration?
- 61. What is appender?
- 62. What is log pattern?
- 63. How to configure logging in Spring Boot?
- 64. Difference between System.out and logging?

6. JIRA

- 65. What is JIRA?
- 66. Why JIRA is used?
- 67. What is issue in JIRA?
- 68. What are issue types?
- 69. What is workflow in JIRA?
- 70. What is sprint?
- 71. What is backlog?
- 72. What is story, task, bug?
- 73. What is epic?
- 74. What is Kanban vs Scrum?
- 75. What is velocity?
- 76. What is burndown chart?

7. Docker

- 77. What is Docker?
- 78. What is containerization?
- 79. Difference between container and virtual machine?
- 80. What is Docker image?

81. What is Docker container?
82. What is Dockerfile?
83. What is Docker Hub?
84. What is docker build?
85. What is docker run?
86. What is docker-compose?
87. What is port mapping?
88. What is volume in Docker?
89. What is container lifecycle?
90. Advantages of Docker?
91. How Docker helps in microservices?

8. Scenario-Based Questions

92. How Maven resolves dependency conflicts?
93. When would you use Gradle over Maven?
94. How do you manage large projects using Git?
95. How to resolve merge conflict in real scenario?
96. How logging helps in production debugging?
97. How JIRA is used in agile project?
98. How to deploy application using Docker?
99. How to run multiple services using Docker Compose?

9. Coding / Practical Questions

100. Write Maven dependency example.
101. Write Gradle dependency example.
102. Create Lombok-based entity class.
103. Git commands for branching and merging.
104. Configure logging in Spring Boot.
105. Write Dockerfile for Spring Boot app.